



UNIVERSITY of the
WESTERN CAPE

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Practical Assignment 1 [50 marks] Deadline: 6th August 2025 at 22H00

This practical aims to help you understand the nature of file systems as used in the early information systems, with emphasis on [structural dependence](#) and [data dependence](#) inherent in file systems. The end goal is to let you experience the "fun" in the job of a data processing specialist when getting answers to the simplest questions on data in the early days. This further clarifies why databases are desirable as a better way of data management in the present times.

You have been provided with a file, which is a file system database that contains thousands of records about different countries.

Your task is to:

1. Write a Python (or Java) program that takes in the file and answers the following questions:
 - a) How many country names end with the letter 'a'?
 - b) List the five cities that have the highest City population.
 - c) List the five countries that have the largest LandMass.
 - d) How many countries gained independence between the years 1960 and 1980 (inclusive)?
 - e) Which countries gained independence between the years 1830 and 1850 (inclusive)?
 - f) List the five African countries that have the highest life expectancy.
 - g) Which are the 5 most commonly spoken languages in the world?
 - h) List the country names that end with the letter 'a', without any repetitions.

NB: Create a second file (named file2) and write the results of the questions from 1 (a to h) in this second file.

Hence, once your program is compiled, it should create this second file with the results displayed in the file. The answers should be displayed in the following format:

Question a:

[Answer]

Question b:

[Answer]

2. Write a report (not more than 8 pages) of how you went about your solution and answer the additional questions in section 3 of the report template provided.

3. Presentation: Each member of the group must indicate how/what they have contributed to the assignment in the report. Furthermore, a physical presentation would be required for final grading. The presentation would take place on 8 August during the practical slots. Details about the venue for the presentation will be communicated in due course.

SUBMISSION AND GRADING

Please submit a (zipped) folder named **TeamName_Practical_1.zip**, which contains your Java/Python implementation (**only JAVA code not .class files**). Also, submit a report (in .PDF format). Please comment your teamName, student number, file name and DB_Practical 1 in the Java/Python file.

Marks will be allocated as follows:

S/N	Task	Points	Weight
1	Code	16	32%
2	Report	20	40%
3	Presentation	14	28%
	Total	50	100%

INSTRUCTIONS

1. Group Formation

- Work in groups of 5–6 members. (NB: Not less than 5, and not more than 6 members are allowed in a group)
- **The group leader must email the list of members** (full names and student numbers) to: csc312@uwc.ac.za latest by **Thursday, 31 July**.
- Email Subject Line: CSC312 DB Practical – (Team Name)
- *Note only use the same group name as the SE project if it's the same members*

2. Assignment Structure & Marks

Q1: Code Implementation (16 marks)

- Correctness (meets functional requirements)
- Efficiency (optimized performance)
- Readability (clear structure, comments, naming)
- Answers all the questions and handles errors

Q2: Written Report (20 marks)

- Followed the template and formatting requirements (Times New Roman 12 pt)
- Clarity and completeness
- Technical depth
- Answered the questions with clarity and appropriate use of terminology

Q3: Presentation (14 marks)

- Clear delivery
- Equal participation of members
- Presentation is a good reflection of what is in the report and code
- Each team will present for **7 minutes**

NB:

1. Since this assignment will be done in groups, please note that there will be no extension. Kindly manage your tasks effectively within your groups.
2. Any group that fails to submit its name and details of members to csc312@uwc.ac.za by 31 July will be automatically grouped because the group submission link needs to be created on ikamva ASAP.